The rate of reaction

Learning outcomes

By the end of this section you should be able to:

- Determine rates of reaction from the tangents of graphs of concentration, volume or mass against time.
- Explain how changes in reaction rate can be monitored indirectly by monitoring changes in mass, volume or colour.

In the study of chemical reactions we are mainly concerned with the rate of reaction. Different reactions proceed at different rates. How can we tell whether a reaction is fast or slow? Generally we measure how fast the reactants disappear or how fast products form, over a given period of time. Fortunately, we do not have to try to count the actual number of molecules of the reactants or products. We can measure a quantity that is directly related to the number of molecules or their concentration. For example, for certain reactions where gases are produced the volume of gas produced over a given time can be measured easily and gives us the rate of the reaction.

The rate of reaction is expressed as the change in the concentration of the product or reactant per unit time.

$$\text{rate} = \frac{\text{change in concentration of reactants or products}}{\text{time}}$$

We know that as a reaction proceeds, reactant concentration decreases and product concentration increases. Watch the two parts of the same video in **Interactive 1** (left video first and use the full screen function if required) that show the change in concentration of reactants and products in a chemical reaction.

Interactive 1. Change in concentration of reactants and products in a chemical reaction.

More information for interactive 1

1
00:00:00,240 --> 00:00:03,040
narrator: About chemical kinetics.
2
00:00:04,360 --> 00:00:09,640
Chemical kinetics is the study of rates
of chemical processes.
3
00:00:10,320 --> 00:00:13,320

The rate of reaction or speed of reaction
4
00:00:13,400 --> 00:00:18,080
can be defined as the change
in concentration of reactant or product
5
00:00:18,280 --> 00:00:20,560
in a particular reaction.
6
00:00:20,880 --> 00:00:23,080
Let us consider this
7
00:00:23,000 --> 00:00:27,080
round bottomed flask contains
N number of A molecules
8
00:00:29,320 --> 00:00:32,040
within a particular period of time.
9
00:00:33,080 --> 00:00:37,960
Like, here, T one is the initial time.
10
00:00:40,800 --> 00:00:42,720
After starting the timer,
11
00:00:43,560 --> 00:00:50,200
N number of A molecules
are converted into B molecules.
12
00:00:51,120 --> 00:00:56,920
Now we can see the disappearing
of A molecule that is reactant molecule,
13
00:00:57,000 --> 00:01:01,520
and appearing of B molecules,
that is product molecules.
14
00:01:04,080 --> 00:01:09,200
At the final time, T2,
all the B molecules are formed
15
00:01:10,040 --> 00:01:14,160
and settle down
at the bottom of the RB flask.
16
00:01:14,240 --> 00:01:16,080
That is a round bottomed flask.
17
00:01:18,040 --> 00:01:22,080
Now we can write this reaction
as a chemical equation
18
00:01:25,400 --> 00:01:31,080
A is disappearing and B is appearing.
19
00:01:32,120 --> 00:01:34,880
Now we can write the rate.
20
00:01:35,400 --> 00:01:40,520
Rate is equal to change in concentration
divided by time taken.
21
00:01:43,080 --> 00:01:46,680
Mathematically it can be written as
22
00:01:48,320 --> 00:01:50,520
negative sign of delta E.
23
00:01:50,840 --> 00:01:53,600
Delta E is the change in concentration.
24
00:01:54,080 --> 00:01:57,920
Concentration is always represented
within the square bracket
25
00:01:58,000 --> 00:01:59,720
divided by delta T.
26
00:01:59,840 --> 00:02:02,600
Now we write the full form of delta E

27
00:02:05,120 --> 00:02:07,720
concentration of A a T2 time
28
00:02:07,800 --> 00:02:13,120
minus concentration of A at T1 time
whole divided by T2 minus T1.
29
00:02:14,080 --> 00:02:17,320
We can write this rate of
30
00:02:19,560 --> 00:02:23,480
reaction in the differential format
31
00:02:24,000 --> 00:02:27,240
R is equal to minus D of A dividend by DT
32
00:02:27,320 --> 00:02:31,200
or otherwise plus D of B divided by DT.
33
00:02:32,080 --> 00:02:33,720
Here, a negative sign indicates
34
00:02:33,800 --> 00:02:38,640
a decreasing in concentration
of reactant molecule .
35
00:02:38,720 --> 00:02:44,760
Positive sign indicates increasing
the concentration of product molecule.
36
00:02:45,320 --> 00:02:49,200
Now we draw a graph
between concentration versus time.
37
00:02:50,120 --> 00:02:56,440
In this graph, we can see
how the concentration of A
38
00:02:56,520 --> 00:03:03,040
decreases and the concentration
of B increases with the increasing time.
39
00:03:04,040 --> 00:03:06,000
In the next video, we will discuss
40
00:03:06,080 --> 00:03:11,240
how to write the rate of the reaction
for the stoichiometric equation.
41
00:03:11,320 --> 00:03:12,120
Thank you.

If we were to measure the rate of reaction as a change in the concentration of the products we could write the rate as shown in **Table 1.**

Table 1. Equation for rate as a change in the concentration of the products.

Equation	Symbols	Units
$\text{rate} = \frac{\Delta[P]}{\Delta t}$	rate	moles per cubic decimetre per second (mol dm^{-3} s^{-1})
	$\Delta[P]$ = the change in concentration of the products	moles per cubic decimetre (mol dm^{-3})
	Δt = the change in time	seconds (s)

Recall from section S1.4.5a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/molar-concentration-of-a-solution-id-44263/) that the square brackets mean concentration.

Similarly, if we were to measure the change in concentration of reactants the equation can be written as shown in **Table 2**.

Table 2. Equation for rate as a change in the concentration of the reactants.

Equation	Symbols	Units
$\text{rate} = -\frac{\Delta[R]}{\Delta t}$	rate	moles per cubic decimetre per second (mol dm^{-3} s^{-1})
	$\Delta[R]$ = the change in concentration of the reactants	moles per cubic decimetre (mol dm^{-3})
	Δt = the change in time	seconds (s)

Study skills

The negative sign in the rate equation in Table 2 indicates that the concentration of the reactants is decreasing but by convention, the rate of a reaction is always taken as a positive value.

Measuring rates of reaction

As you know, the rate of a reaction can be determined experimentally by measuring the change in concentration of the reactants or products. Whether we choose to measure the change in concentration of products or reactants depends on the nature of the reaction. Here are some of the techniques that can be used.

Measuring the change in volume of a gas

The reaction rate for the reaction between an acid and a metal to produce hydrogen gas can be found by measuring the volume of hydrogen produced over regular intervals. This will enable us to draw a graph of volume vs. time.

$$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$$

Gas collection can be done in several ways. A gas syringe can be used to collect the gas. The gas syringe is specially designed with a glass barrel that moves outward as it fills with gas. As the syringe is calibrated, the volume of gas can be measured directly and at the same time a stopwatch can be used to measure the time (**Figure 1**).

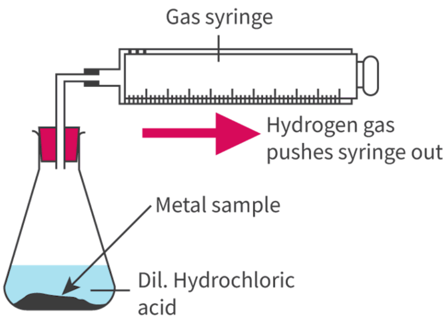


Figure 1. Collection of hydrogen gas using the gas syringe.

More information for figure 1

The diagram illustrates the process of collecting hydrogen gas using a gas syringe. It features a setup with a flask that contains a metal sample and diluted hydrochloric acid. A line connects the flask to a gas syringe, which is shown extending as it fills with hydrogen gas. An arrow points from the hydrogen gas, labeled as pushing the syringe outward. The design shows how the gas syringe is calibrated to measure the volume of hydrogen gas collected during the reaction, where the metal interacts with the acid to produce the gas.

[Generated by AI]

Alternatively, the data logging equipment shown in **Figure 2** can be used to measure the quantity of gas produced by using a gas pressure sensor. The volume of gas produced is directly proportional to the pressure of the gas (**Figure 3**). Pressure changes are particularly useful for reactions in which both reactants and products are gaseous.

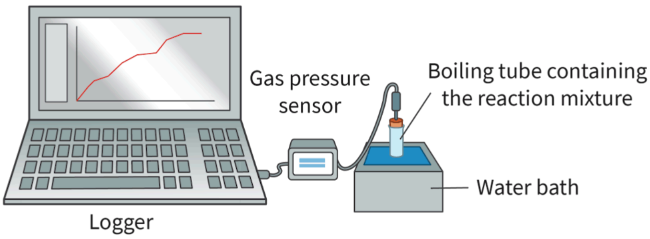


Figure 2. A gas pressure data logger.

More information for figure 2

The image is an illustration showing a gas pressure sensor setup connected to a laptop acting as a data logger. The system includes a boiling tube containing the reaction mixture submerged in a water bath. A gas pressure sensor is connected to the boiling tube, which is in turn connected to the laptop. The screen of the laptop displays a red line graph representing pressure data over time, as the data logger records the changes. Labels in the image identify each component: "Logger" next to the laptop, "Gas pressure sensor" near the sensor apparatus, "Boiling tube containing the reaction mixture" pointing to the tube, and "Water bath" at the base where the tube is situated.

[Generated by AI]

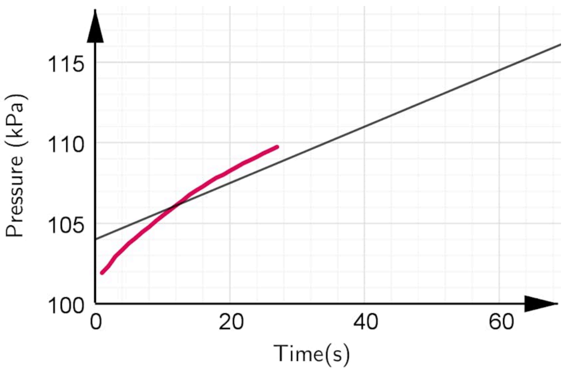


Figure 3. Results of an experiment to determine the rate of reaction by following change in pressure against time using a gas pressure sensor.

More information for figure 3

The graph displays pressure in kilopascals (kPa) on the Y-axis against time in seconds (s) on the X-axis. The Y-axis ranges from 100 to 115 kPa, and the X-axis ranges from 0 to 60 seconds. There are two lines present: one is a red line depicting experimental data, which starts slightly above 100 kPa and rises to just above 110 kPa by approximately 30 seconds, showing a non-linear increase initially and leveling out somewhat. The other line is a straight, black line showing a steady increase, serving as a reference or control.

[Generated by AI]

Measuring the changes in mass

Many reactions in which gases are produced will show a decrease in mass if the gas is removed from the reaction mixture. This change in mass can be measured over time and the rate of reaction determined. The rate of a reaction between a metal and acid producing hydrogen gas can be monitored with this method, as shown in **Figure 4**.

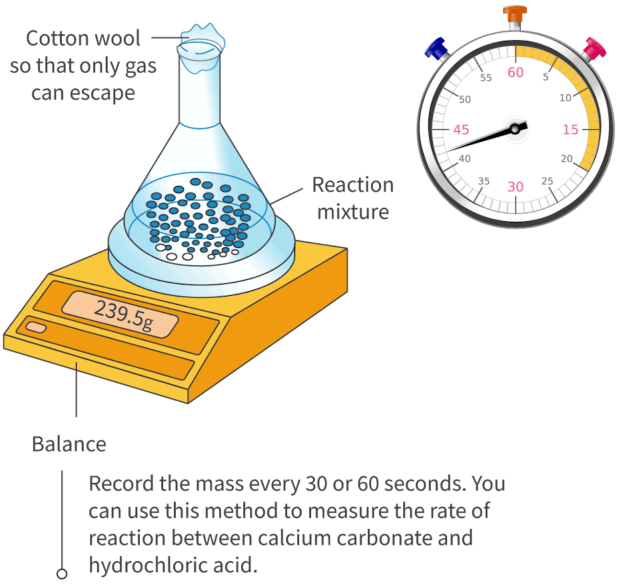


Figure 4. Measuring the rate of reaction by following changes in mass against time.

More information for figure 4

The illustration shows a laboratory setup to measure the rate of reaction through changes in mass over time. A conical flask is placed on an electronic balance that reads 239.5g. The flask contains a reaction mixture represented by small granular particles. It is covered with cotton wool, described as allowing only gas to escape. Next to the balance, there is a large stopwatch displaying time intervals with notable markings at every 5-second increment, highlighting the 30-, 45-, and 60-second points. The text suggests recording the mass every 30 or 60 seconds and notes that this method is used for reactions, such as those between calcium carbonate and hydrochloric acid.

[Generated by AI]

Measuring the changes in colour

A colorimeter or spectrophotometer is a device that measures the amount of light transmitted or absorbed by a coloured solution. Light of specific wavelength is selected and passed through the solution sample and the absorbance (or transmittance) is measured and plotted against time. We know that coloured solutions absorb some parts of the visible light spectrum and transmit others. As the concentration of some solutions increases, they become darker and absorb proportionally more light. This means less light is transmitted.

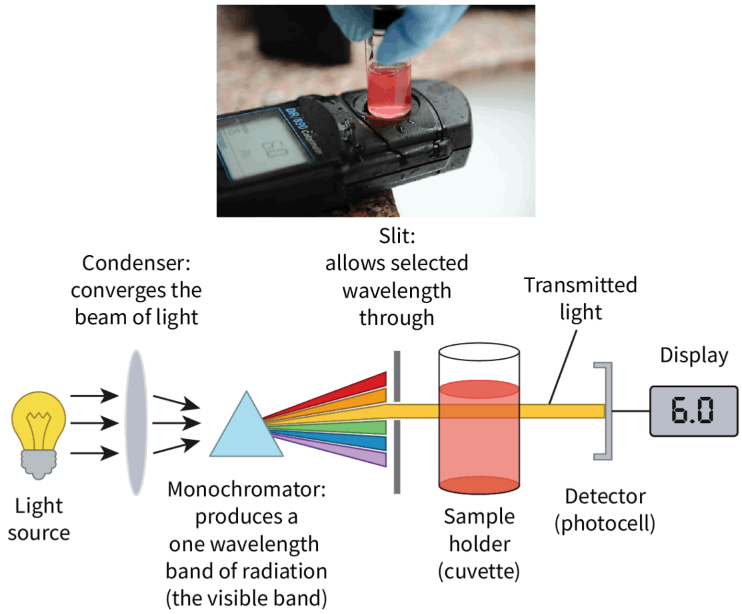


Figure 5. The simplified working of a colorimeter.

More information for figure 5

The image is a diagram illustrating the working components of a colorimeter accompanied by an inset photo of an actual colorimeter in use. It starts with a bulb labeled as 'Light source,' which emits light. This light passes through a condenser labeled 'Condenser: converges the beam of light,' forming a focused beam of light. The beam then enters a monochromator, depicted as a prism, labeled 'Monochromator: produces a one wavelength band of radiation (the visible band).' The monochromator releases a spectrum from which light of a specific wavelength is selected and directed through a slit with the label 'Slit: allows selected wavelength through.' The light then travels through a sample holder labeled 'Sample holder (cuvette),' filled with a colored solution. Next, the light labeled 'Transmitted light' travels to a detector, also referred to as 'photocell,' which measures the light intensity. Finally, a digital output labeled 'Display' shows the measurement as '6.0.' The inset photo above shows the top view of a real colorimeter with a hand placing a sample into it.

[Generated by AI]

The reaction between propanone and iodine starts off with a yellow brownish colour of iodine and gradually fades to colourless. This means we can follow the reaction using colorimetry.

$$CH_3COCH_3(aq) \quad + \quad I_2(aq) \xrightarrow{H^+ \text{ catalyst}} CH_3COCH_2I(aq) \quad + \quad HI(aq) \quad \text{colourless} \qquad \text{(yellow/brov}$$

As the reaction moves towards completion and the solution becomes lighter in colour the absorbance values decrease.

Certain colorimeters can also be attached to data loggers to obtain continuous readings that are then represented in a graphical form. **Video 1** explains how to operate a simple colorimeter in the laboratory.

Video 1. Using a colorimeter.

Other methods for measuring reaction rates

Measuring changes in conductivity

A change in the concentration of ions in solution leads to a change in conductivity. Hence reactions, that involve changes in ion concentration when reactants change into products, can be monitored by taking conductivity measurements (**Figure 6**).

For example,

$$(CH_3)_3CBr(aq) + H_2O(l) \rightarrow (CH_3)_3COH(aq) + H^+(aq) + Br^-(aq)$$

In this reaction, an increase in the concentration of ions in the solution (2 moles of ions on the right-hand side and 0 moles of ions on the left-hand side) will manifest as an increase in conductivity during the progress of the reaction.

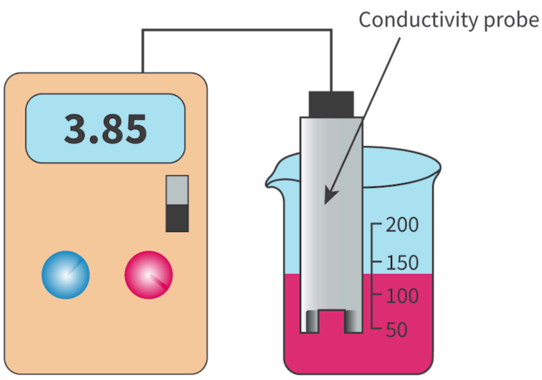


Figure 6. An electrical conductivity meter.

More information for figure 6

The image depicts an electrical conductivity meter connected to a probe. On the left, there is a digital device displaying the value 3.85. The device has two colored buttons below the display, one blue and one pink, and a gray switch on the right side. A wire leads from this device to a conductivity probe inserted into a transparent beaker filled with pink liquid. The beaker has measurement markings on the side, indicating levels at 50, 100, 150, and 200 units. The probe has an arrow pointing downward into the liquid, labeled 'Conductivity probe.' The image illustrates a setup for measuring the conductivity of a liquid, likely related to a chemical reaction where ion increase is measured.

[Generated by AI]

Measuring the rate of formation of precipitate

The sodium thiosulfate and hydrochloric acid reaction produces a pale yellow precipitate of sulfur over a period of time which results in the clear solution turning opaque.

$$Na_2S_2O3(aq) + 2HCl(aq) \rightarrow 2NaCl(aq) + SO_2(g) + S(s) + H_2O(l)$$

The progress of this reaction can be followed by measuring the time taken for the solution to turn opaque, so that a cross viewed through the solution is no longer visible (**Figure 7**).

A video of the 'disappearing cross' reaction can be seen in the next section R2.2.2–3a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/factors-affecting-the-rate-of-reaction-id-44714/), where the effect of concentration on the reaction is investigated.

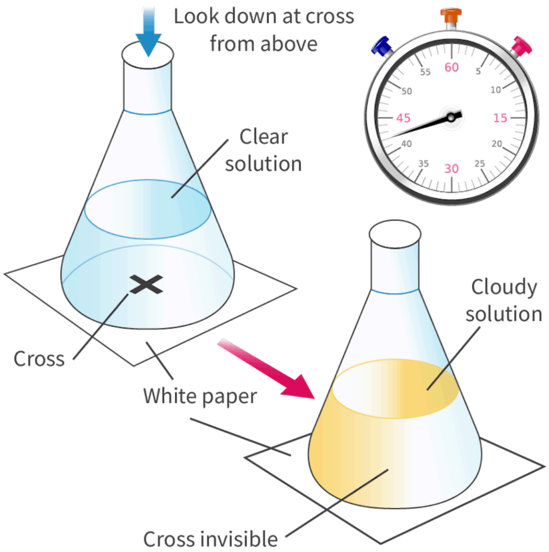


Figure 7. Recording the increase in opaqueness during a reaction.

More information for figure 7

The image is a diagram illustrating the 'disappearing cross' experiment. It contains two conical flasks positioned over a cross drawn on paper. The first flask shows a clear solution, through which the cross is visible. An arrow indicates the progression from the first to the second flask, which contains a cloudy solution, obscuring the cross. A stopwatch is displayed beside the flasks, marking time intervals. Labels identify the 'clear solution', 'cloudy solution', and the 'cross invisible' condition as the reaction progresses.

[Generated by AI]

Measuring the rate of clock reactions

A clock reaction generally involves a series of chemical reactions where an additional reaction can only occur after the reaction(s) under study is/are complete. This clock reaction will involve a very rapid colour change or formation of a precipitate. By measuring the time taken for the visible change of the final reaction in the series to occur or repeat itself, chemists can measure the rate of the reaction under study.

For example, in the iodine clock reaction two colourless solutions are mixed, one containing hydrogen peroxide and sulfuric acid and the other containing potassium iodide, sodium thiosulfate and starch. At first there is no visible reaction but only when the first two reactions are completed, can the final reaction occur, and the mixture turns dark blue (**Figure 8**).

In the first set of reactions iodine is produced

$$H_2O_2(aq) + 2I^-(aq) + 2H^+(aq) \rightarrow I_2(aq) + 2H_2O(l) \quad \text{no visible changes}$$

This iodine is then used up by the thiosulfate.

$$2S_2O_3^{2-}(aq) + I_2(aq) \rightarrow 2I^-(aq) + S_4O_6^{2-}(aq) \quad \text{no visible changes}$$

The amounts chosen are such that as soon as the iodine is in excess (and the thiosulfate is limiting) the iodine reacts with the starch to form a blue-black complex.

$$I_2 + \text{starch} \rightarrow \text{dark blue iodine-starch complex}$$

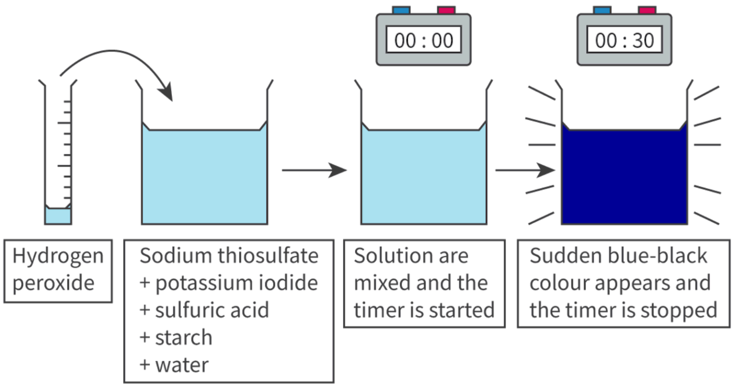


Figure 8. The iodine clock reaction.

More information for figure 8

The image is a diagram depicting the iodine clock reaction process in four steps.

1. **Hydrogen Peroxide Introduction**: A beaker is shown with a label "Hydrogen peroxide." This is added to another solution.
2. **Solution Mixture**: A new beaker, now containing a blue liquid, is labeled with its components: sodium thiosulfate, potassium iodide, sulfuric acid, starch, and water.
3. **Timer Started**: The mixture is stirred, initiating the reaction, marked by a digital timer displaying 00:00.
4. **Color Change and Timer Stop:** The solution changes rapidly to a blue-black color, indicated by a visual cue from a different beaker, and the timer shows 00:30. Text states that the color change is sudden, signaling the reaction completion.

[Generated by AI]

Study skills

Spot reactions where both reactants are aqueous, and a product is produced in the form of a precipitate. This is shown by the state symbol for a solid 's' next to the product, as in the production of sulfur in the sodium thiosulfate and hydrochloric acid reaction. A precipitate is always opaque and the rate of such reactions can be measured by monitoring changes in solution clarity.

Calculating reaction rates from graphs

The graph in **Figure 9** shows the change in concentration of products for two different reactions. We can see that products are formed more quickly in the blue curve as compared to the red curve. That is, the curve is steeper. Hence the steepness of the curve is an indication of 'how quickly' the reaction proceeds. The steepness of the curve can be measured mathematically by calculating the gradient or slope of the line. The gradient of a line is calculated by measuring the change in concentration over time and is therefore a measure of the rate of the reaction.

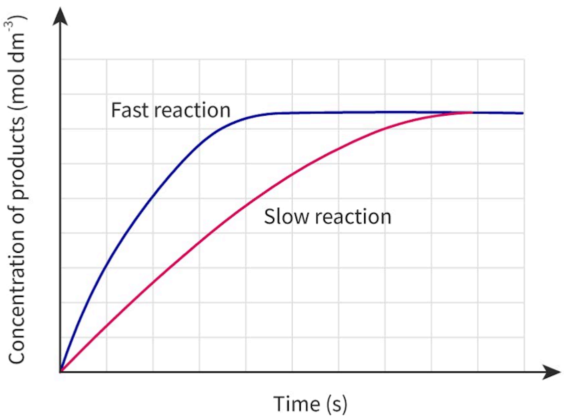


Figure 9. A graph comparing rates of reaction.

More information for figure 9

The image is a graph displaying the rates of reactions by comparing two different reaction curves over time. The X-axis is labeled "Time (s)" and the Y-axis is labeled "Concentration of products (mol dm^-3)." There are two curves displayed on the graph: a blue curve and a red curve. The blue curve, labeled "Fast reaction," shows a steep increase in the concentration of products, indicating a rapid reaction rate. In contrast, the red curve, labeled "Slow reaction," shows a more gradual increase in concentration, representing a slower reaction rate. The steepness of each curve signifies the speed of the respective reaction. As time progresses, both curves plateau, indicating that the reactions have reached completion, with the fast reaction achieving this state quicker. The grid lines on the graph provide a visual reference for estimating values but are not numerically labeled.

[Generated by AI]

Calculating the instantaneous rate of a reaction

As reactions proceed, the rate decreases over time until one or more of the reactants is used up and the rate becomes zero. This means the rate of reaction or gradient keeps changing so instantaneous rate is often taken which is the rate at a particular time. This can be done by drawing a tangent to the graph at that particular time. **Figure 10** shows how the instantaneous rate can be measured at 12 seconds.

Calculating the initial rate of a reaction

The shape of a typical rate curve (**Figure 10**) shows that the rate of reaction is greatest at the start and slows down as the reaction proceeds. The initial rate measures the rate of reaction at the start of the reaction. Initial rates data is important when comparing the effect of various factors on the reaction rates.

The initial rate of reaction can be calculated by drawing a tangent at 0 s and calculating the gradient as shown.

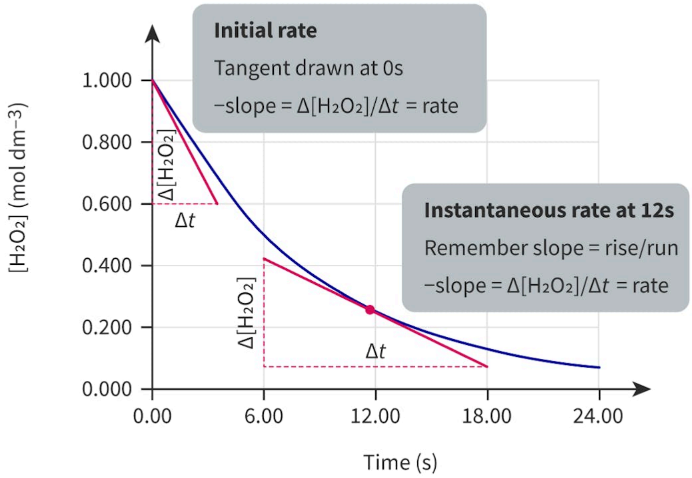


Figure 10. Graph of concentration vs. time for the decomposition of hydrogen peroxide. Tangents are shown at t = 0 s and t = 12 s.

More information for figure 10

The graph displays the concentration of hydrogen peroxide [H2O2], measured in mol dm^-3, on the Y-axis versus time in seconds on the X-axis. The Y-axis ranges from 0.000 to 1.000 mol dm^-3, while the X-axis spans from 0 to 24 seconds. The curve represents the decrease in concentration over time, showcasing the decomposition process of hydrogen peroxide. Two tangents are drawn: one at time 0 seconds and another at time 12 seconds. The tangent at t=0 seconds intersects the graph at 1.000 mol dm^-3 and corresponds with a slope given by the rate of reaction formula Δ[H2O2]/Δt. Similarly, the tangent at t=12 seconds relates to the instantaneous rate of reaction at that time, emphasizing that the slope reflects the rate as -Δ[H2O2]/Δt. The graph visually aids in understanding the initial and instantaneous rates of decomposition by illustrating the changes in slopes and concentration over time.

[Generated by AI]

Calculating the average rate of a reaction

At times when reactions proceed very quickly it is not easy to measure the instantaneous rate of the reaction and the average rate is taken. The average rate of reaction is an average of all the instantaneous reaction rates over a given period of time. The average rate of the reaction can be calculated by drawing a secant to the rate curve and measuring the gradient, as in **Figure 11**.

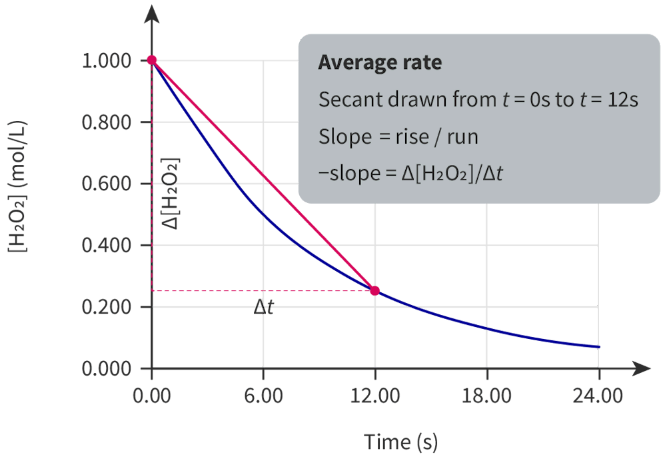


Figure 11. A graph of concentration vs. time for the decomposition of hydrogen peroxide. Secant shown from t = 0 s to t = 12 s, to calculate the average rate in the first 12 s.

More information for figure 11

The image is a graph representing the concentration of hydrogen peroxide ([H2O2]) over time. The X-axis is labeled "Time (s)" and ranges from 0 to 24 seconds, while the Y-axis is labeled "[H2O2] (mol dm^−3)" ranging from 0.000 to 1.000. A curve plotted in blue shows the decomposition of hydrogen peroxide over time, descending from the top left (1.000 mol dm^−3 at 0 seconds) towards the x-axis, indicating a decrease in concentration as time progresses. A red secant line is drawn from the point where time equals 0 to the point at 12 seconds to calculate the average rate of change in this interval. The gray box within the graph provides a description of the average rate, showing that the slope of the secant is calculated by the formula: Slope = rise/run = Δ[H2O2]/Δt.

[Generated by AI]

Worked example 1

Figure 12 shows the volume of gas produced during a reaction.

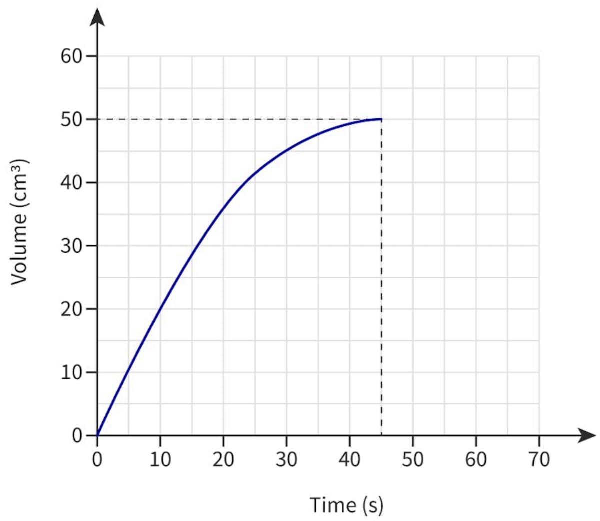


Figure 12. Volume vs. time graph for a reaction.

More information for figure 12

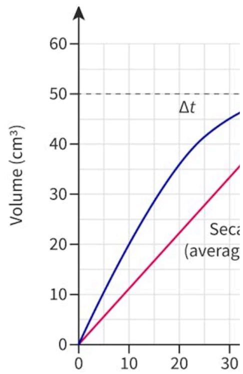
The image is a graph illustrating the relationship between volume and time for a chemical reaction. The X-axis represents time in seconds, ranging from 0 to 70 seconds, marked at intervals of 10 seconds. The Y-axis denotes volume in cubic centimeters (cm^3), ranging from 0 to 60 cm^3, with an interval of 10 cm^3. A curve on the graph shows an increase in volume over time, starting at the origin (0,0) and rising sharply upwards to approach 50 cm^3 by the time it reaches approximately 50 seconds. The curve then levels out, showing a plateau as it reaches the maximum volume at around 50 cm^3. Dashed lines extending horizontally and vertically from the curve to the axes indicate this plateau point.

[Generated by AI]

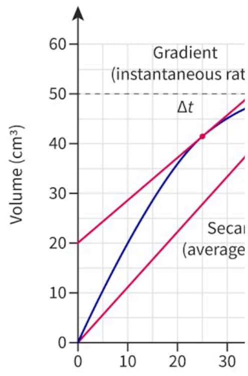
Find

1. the average rate of the reaction from the graph,
2. the instantaneous rate at 25 s.

1. Calculating the average rate of reaction.

Solution steps	Calculations
Step 1: Draw the secant line.	
Step 2: Write out the required formula.	Find the gradient of the seca $\text{gradient} = \frac{\text{rise}}{\text{run}}$
Step 3: Write out the known values.	$\text{rise} = (50\text{-}0) = 50\ \text{cm}^3$ $\text{run} = (45\text{-}0) = 45\ \text{s}$
Step 4: Substitute the values given.	$= \frac{50}{45}$
Step 5: State the answer with appropriate units and the number of significant figures used in rounding.	$= 1.1\ \text{cm}^3\ \text{s}^{-1}\ (2\ \text{s.f.})$ The unit for rate of reaction i This is acceptable when the r the volume of gas produced

2. Calculating the instantaeous rate of reaction at 25 s.

Solution steps	Calculations
Step 1: Draw the tangent line.	
Step 2: Write out the required formula.	Find gradient of tangent line a $\text{gradient} = \frac{\text{rise}}{\text{run}}$
Step 3: Write out the known values.	$\text{rise} = (57\text{-}20) = 37\ \text{cm}^3$ $\text{run}\ (43\text{-}0) = 43\ \text{s}$
Step 4: Substitute the values given.	$= \frac{37}{43}$
Step 5: State the answer with appropriate units and the number of significant figures used in rounding.	$= 0.86\ \text{cm}^3\ \text{s}^{-1}\ (2\ \text{s.f.})$

Creativity, activity, service

Strand: Service
Learning outcomes:

- Demonstrate engagement with issues of global significance
- Recognise and consider the ethics of choices and actions

The average person produces approximately 2 kg of solid waste every day. While some of this waste can be recycled, about half of it ends up in landfill sites (https://www.eastsussex.gov.uk/rubbish-recycling/how-we-manage-our-waste/landfill-sites/what-is-a-landfill-site#:~:text=A%20landfill%20site%20is%20an,unwanted%20hole%20in%20the%20ground.). Once in the landfill site, this waste decomposes but only very slowly depending on factors such as the amount of oxygen available, the temperature and the moisture. The decomposition of this waste can also produce greenhouse gases (https://www.epa.gov/lmop/basic-information-about-landfill-gas) such as methane, CH_4.

Think about what you can do to reduce the amount of waste produced by your school or in your local community — what can you do to reduce the amount of waste that ends up in landfill? Perhaps you could start by raising awareness of issues such as these as a possible CAS experience?

Activity

- **IB learner profile attribute:**
  - Knowledgeable
  - Inquires
  - Thinkers
  - Reflective
- **Approaches to learning:**
  - Thinking skills — Providing a reasoned argument to support a conclusion; Applying key ideas and facts in new contexts
  - Communication skills — Using terminologies, symbols and communication conventions consistently and correctly
  - Social skills — Working collaboratively to achieve a common goal
- **Time required to complete activity:** 25—30 minutes
- **Activity type:** Individual/pair activity

Work in pairs to complete the following tasks:

1. For the reaction between marble chips (calcium carbonate) and hydrochloric acid the rate of reaction can be measured in two ways, illustrated in **Figure 13.**

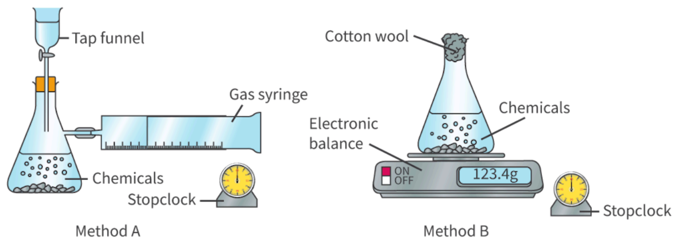


Figure 13. Two methods for measuring rate of reaction between marble chips and dilute hydrochloric acid.

More information for figure 13

The image shows two different setups for measuring the reaction rate between marble chips and dilute hydrochloric acid. On the left, Method A involves a conical flask with chemicals labeled inside. A tap funnel is placed at the top of the flask for acid addition, and there's a gas syringe connected to the side to measure gas output. A stopclock is placed nearby to track the time of reaction. On the right, Method B uses a conical flask containing similar chemicals and has cotton wool plugged at the mouth to prevent spillage while allowing gas escape. This flask is placed on an electronic balance displaying 123.4g to measure mass change. A stopclock is also present to record the time. Each method aims to observe the rate at which carbon dioxide is produced when marble chips react with the acid.

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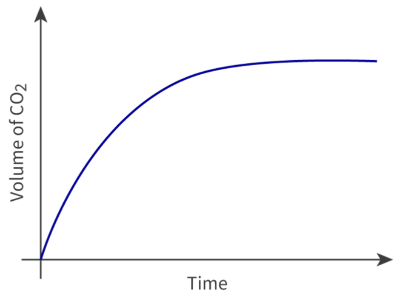
(a) Sketch a graph of the carbon dioxide produced against time in Method A.
(b) Explain why it is important to have a tap funnel that allows acid to be added when needed?
(c) Sketch the graph you would obtain if you plotted the mass of the flask, shown by the electronic balance, against time in method B.
(d) Sketch the graph you would expect if you plotted the mass of carbon dioxide given off against time in method B.
(e) Explain why cotton wool is plugged into the mouth of the conical flask in Method B.

1. You are provided with a table containing six reactions. Five possible options of experimental techniques that may be used to measure the rate of each reaction, are given at the bottom of the sheet. Download the worksheet and complete the table by writing the correct experimental technique next to its respective reaction. More than one option can be

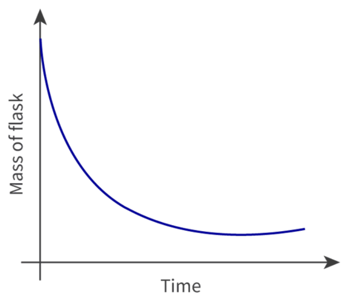
chosen for a reaction.

Worksheet
(https://d3vrb2m3yrmyfi.cloudfront.net/media/edusys_2/content_uploads/Che

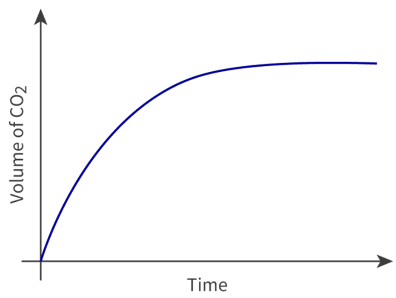
1. (a)



(b) The funnel allows hydrochloric acid to be added when the entire setup is sealed and ready to go. If we were to add the acid by opening the flask, this may result in the loss of gas as soon as it is produced, leading to an error in measuring the initial rate of the reaction.
(c)



(d) Same as in part (a).



(e) The cotton wool prevents droplets of solution, that may be thrown out of the solution by the effervescence, from escaping. At the same time, it allows the carbon dioxide to pass through.

1.

Reaction	Reaction equation	Method to monitor rate
1	$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$	Change in pressure or volume
2	$10FeSO_4$ <pale green> + $2KMnO_4$ <purple> + $8H_2SO_4 \rightarrow K_2SO_4$ + $2MnSO_4$ <pale pink> + $5Fe(SO_4)_3$ <yellow brown> + $8H_2O$	Colorimetry
3	$C_2H_4 + Br_2$ <red brown> $\rightarrow C_2H_4Br_2$ <colourless>	Colorimetry
4	$BrO_3^- + 5Br^- + 6H^+ \rightarrow 3Br_2(l)$ <brown> + $3H_2O(l)$	Colorimetry/ Conductivity
5	$NH_4Cl + NaOH \rightarrow NH_3 + NaCl + H_2O$	Change in pressure or volume / Change in mass
6	$5H_2O_2(aq) + 2HIO_3(aq)$ <yellow> $\rightarrow 5O_2(g) + I_2(aq)$ <blue black> + $6H_2O(l)$ (colour oscillates between yellow and blue black)	Colorimetry/ Conductivity

Experimental techniques for measuring the rate of reaction:
- Colorimetry
- Change in pressure or volume
- Change in mass
- Conductivity
- Clock reaction

1. **Reaction 1**
   Change in pressure or volume can be used as both the reactants and products are gaseous.

   Reaction 2
   Colorimetry can be used to monitor the changes in colour. As the reaction starts the reaction mixture is predominantly purple and this fades to a very pale brown colour as the reaction proceeds.

   Reaction 3
   Changes in colour can be monitored using a colorimeter.

   Reaction 4
   Colorimetry can be used to follow the appearance of the brown colour.

   There are 12 moles of ions on the right side of the equation (reactant side) and no ions on the left side, making it easy to monitor the rate of the reaction using conductivity values.

   Reaction 5
   Both change in pressure or volume and change in mass can be used as ammonia gas is produced.

   Reaction 6
   This reaction, the Briggs Rauscher reaction, is an oscillating reaction that involves a colour change from pale yellow to blue black and then back to yellow again. These changes can be monitored using colorimetry or can be measured as a clock reaction using a stopwatch to measure the periodic change in colour.

   There are no ions on the product side, only oxygen gas, aqueous iodine and water but the reactant HIO_3 exists as H^+ and IO_3^- ions in solution. The decrease in ions from left to right can be monitored by measuring the conductivity of the reaction mixture.

The concept diagram in **Figure 14** links key concepts for measuring reaction rates.

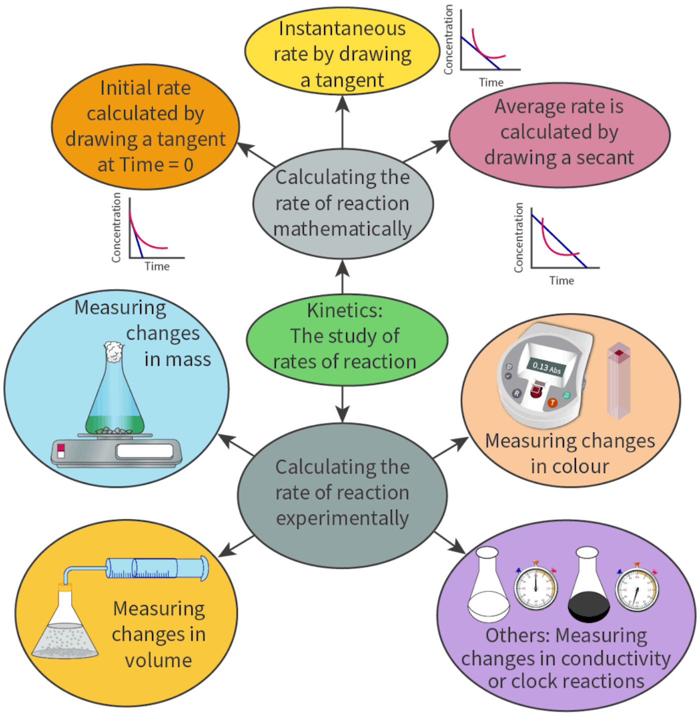


Figure 14. Measuring the rate of reaction.

More information for figure 14

The diagram illustrates the processes and concepts involved in measuring reaction rates. At the center, a gray oval labeled "Calculating the rate of reaction mathematically" connects to multiple surrounding elements.

To the top, a yellow oval states "Instantaneous rate by drawing a tangent," with a small graph showing a curve with tangent. On the top left, an orange oval says "Initial rate calculated by drawing a tangent at Time = 0," alongside a curve graph emphasizing tangency. Top right, a purplish pink oval notes "Average rate is calculated by drawing a secant," with a corresponding curve and secant line on the graph.

Directly below the central oval, a green oval reads "Kinetics: The study of rates of reaction."

On the bottom left, a light blue circle shows an image of a flask placed on a scale with the text "Measuring changes in mass." Further below, a yellow oval labeled "Measuring changes in volume" depicts a syringe setup linked to a flask.

On the bottom right, an orange oval with a spectrophotometer illustration is labeled "Measuring changes in colour." Lastly, on the lower right, a purple circle labeled "Others: Measuring changes in conductivity or clock reactions" depicts flasks and stopwatches.

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